

Kairui Hu

+44 7930391863 | +86 18581030973 | kairui.hu26@imperial.ac.uk | [karryh26.github.io](https://github.com/karryh26)

EDUCATION

Imperial College London

Bachelor and Master in Electrical and Electronic Engineering

Sep. 2026 – Jun. 2030(Expected)

London, UK

EXPERIENCE

Research Assistant in Embodiment Intelligence

Shanghai Innovation Institute

Aug. 2026 – Present(Remote)

Shanghai, China

- Developed a highly efficient algorithm for action selection of world model
- Proposed a mechanism to detect and mend the internal error of prediction model
- Constructed a series of deployed tests for a single flexible thermoelectric device

Internship in Autonomous Driving Algorithm

Relaxing Technology

Jun. 2026 – Aug. 2026

Chongqing, China

- Improved lane detection function under extreme lighting scenario

PROJECTS

Improvement of Adaptive Action-Value Estimation in WM

Aug. 2026 – Present

- Formulated an efficient planning framework that distinguishes candidate action sequences by their expected task value under a stochastic generative world model.
- Proposed a two-fidelity action-value estimator that uses shared randomness and reference caching to reduce estimation variance and computation.
- Designed a confidence-guided planner that allocates computation to ambiguous value comparisons, eliminates clearly inferior actions.

PUBLICATIONS

CoPASS

Anonymous authors

ICRA 2027, Under Review

- Introduces CoPASS, a flexible thermoelectric system combining co-located Peltier heating and Seebeck sensing through time-division multiplexing.
- Demonstrates repeatable recovery responses after active thermal excitation in controlled-contact experiments.
- Achieves 100% classification of PMMA, glass, and stainless steel contacts using three thermal-response features and leave-one-block-out validation.
- Validates integration of the same device and protocol in robotic grasping experiments.

TECHNICAL SKILLS

Robotics & Middleware: ROS 2, AgileX NERO, Pico 4 Ultra, inverse dynamics, teleoperation

Languages: Python, C++, Java

Developer Tools: Git, VS Code, PyCharm, Linux/Ubuntu, LaTeX

Libraries: NumPy, pandas, Matplotlib

AI-Assisted Development: ChatGPT, Claude Code